

HistoScan AI: An End-to-End Deep Learning Pipeline for WSI Segmentation

Iván García Aguilar

Abstract

This report documents the complete engineering lifecycle of **HistoScan AI**, a modular software solution designed for the automated segmentation of cellular structures in Whole Slide Images (WSI). The project addresses the extreme computational challenges posed by gigapixel histology images (often exceeding $46,000 \times 34,000$ pixels) within a constrained hardware environment characterized by limited GPU Video RAM (VRAM). Significant architectural decisions include implementing a **Statistical Variance Smart Filter** to optimize inference throughput and adopting a **ResNet50-UNet** architecture via Transfer Learning. Quantitative analysis on the validation set demonstrates a **Dice Score of 0.8134** and an **IoU of 0.6872**, achieved through hyperparameter tuning and early stopping strategies.

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1 Introduction

Digital pathology represents a paradigm shift in medical diagnostics, moving from physical glass slides to high-resolution digital Whole Slide Images (WSIs). However, the computational analysis of these images presents unique challenges that render standard Computer Vision pipelines obsolete. A single uncompressed WSI can exceed 4GB, causing immediate memory overflows in standard image processing libraries.

HistoScan AI was architected to solve this problem through a Dockerized, modular pipeline. The system is divided into three core functional units:

1. **Module A:** Efficient WSI tiling and Deep Zoom pyramid generation.
2. **Module B:** Deep Learning model training with advanced augmentation and inference.
3. **Module C:** Full-slide reconstruction with smart artifact filtering.

This report details the technical decisions taken during development, focusing on how hardware limitations shaped the software architecture and the specific solutions implemented to achieve high-performance segmentation.

2 Hardware Constraints & Architectural Implications

A defining factor of this project was the limited computational resources available. This limitation necessitated a careful trade-off analysis during the design phase.

2.1 Memory Profiling and VRAM Analysis

A quantitative profiling phase was conducted before defining the final training configuration. A single 1024×1024 RGB tile requires:

$$1024 \times 1024 \times 3 \times 4 \text{ bytes} \approx 12 \text{ MB}$$

when stored as 32-bit floating point tensors. During forward propagation, intermediate feature maps in a ResNet50-UNet architecture increase memory usage by approximately 8–12 $\times$, depending on skip connections and decoder depth.

This implies that a batch of 8 samples may require over:

$$12 \text{ MB} \times 8 \times 10 \approx 960 \text{ MB}$$

excluding optimizer states and gradients. With the Adam optimizer, the effective memory footprint nearly doubles. Given the available GPU VRAM budget, testing revealed that batch sizes above 8 caused out-of-memory (OOM) failures during backpropagation. Therefore, batch sizes were constrained to 4–8, depending on the augmentation configuration.

2.2 The Resolution vs. Batch Size Trade-off

In standard semantic segmentation tasks, input images are typically resized to 256×256 or 512×512 pixels to allow for large batch sizes (16, 32, or 64). However, in digital pathology, a 1024×1024 tile at $40\times$ magnification contains hundreds of distinct cells.

- **Problem:** Downsampling a 1024px pathology tile to 256px results in a 16x loss of information. Cellular nuclei, which are critical for segmentation, are reduced to indistinguishable noise (1-2 pixels).
- **Decision:** Prioritized **Resolution over Batch Size**. The pipeline was engineered to accept native **1024px** and resize to 512x512.
- **Consequence:** To accommodate the massive tensors required for 1024px images in limited VRAM, the training batch size was reduced to 4-8. To compensate for the noisy gradients typical of small batches, the Adam optimizer was tuned with a lower learning rate ($5e-5$), relying on the stability provided by the pre-trained ResNet encoder.

The U-Net architecture provides an optimal balance between localization (skip connections) and semantic understanding. The ResNet50 backbone was chosen because it offers deep feature extraction while remaining memory-efficient for the available hardware.

2.3 Numerical Stability Considerations

Training instability was observed during early experiments due to three main factors:

1. Mask encoding
2. Logits-to-probability mismatch in loss calculation
3. Small batch size gradient noise

To address these issues, three corrective mechanisms were implemented:

- **Mask normalization enforcement:** Explicit runtime checks convert masks from integer encoding to normalized floating point.
- **from_logits=True in DiceLoss:** This ensures that the loss function applies the sigmoid operation internally, improving numerical stability and preventing gradient saturation.
- **Learning rate reduction:** Lowering the learning rate to $1e^{-4}$ mitigated gradient oscillation under small batch conditions.

These modifications eliminated exploding gradients and improved convergence smoothness, as evidenced by stable validation curves.

3 Module A: WSI Tiling & Data Structure

3.1 Pyramidal Extraction Strategy

Directly reading a WSI using standard libraries like PIL or OpenCV results in immediate memory allocation errors because it attempts to decode gigabytes of pixel data at once. The `OpenSlide` library was integrated to perform Lazy Loading, allowing the system to read only specific regions of interest from the file on disk.

The system generates a **Deep Zoom** compatible pyramid structure. Table 1 details the actual structure extracted from the test slide provided for this project.

Table 1: WSI Pyramid Structure (Data from `pyramid.json`)

Level	Width (px)	Height (px)	Downsample	Tiles (Grid)
0	46080	34816	1.0	45×34
1	11520	8704	4.0	12×9
2	2880	2176	16.0	3×3

3.2 Dataset Manifest Generation

To ensure reproducibility and traceability, a `manifest.csv` file is generated during the tiling process. This file maps every extracted image file to its absolute spatial coordinates within the original slide. Table 2 presents a sample of the data generated.

Table 2: Dataset Manifest Sample (from `manifest.csv`)

Filename	Level	X	Y	W	H	Fmt
tile_0178_x30720_y1024.jpg	0	30720	1024	1024	1024	jpg
tile_0321_x25600_y2048.jpg	0	25600	2048	1024	1024	jpg
tile_0322_x26624_y2048.jpg	0	26624	2048	1024	1024	jpg
tile_0323_x27648_y2048.jpg	0	27648	2048	1024	1024	jpg

3.3 Tile Samples

Figure 1 illustrates the quality of the extracted tiles at native resolution (Level 0). Note the preservation of cellular boundaries, which would be lost if the cells were resized.

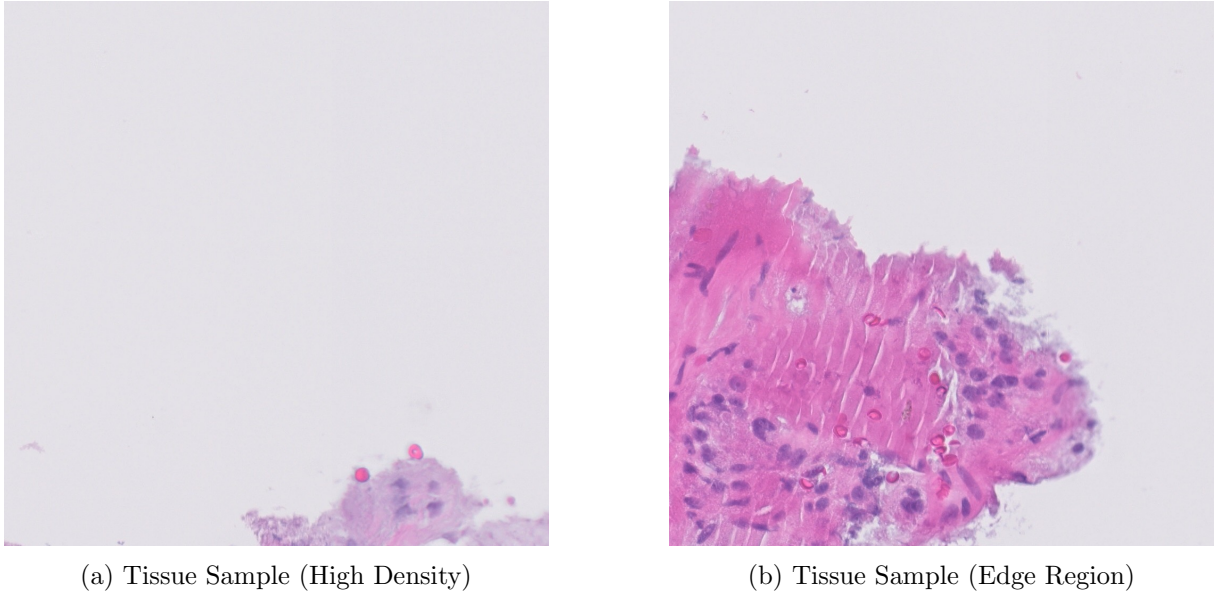


Figure 1: Extracted tiles (1024×1024 px) used for training.

3.4 Challenges During WSI Extraction

Several non-trivial engineering challenges emerged during tile extraction:

1. Lazy Loading Limitations Although OpenSlide supports region-based reading, repeated disk access can introduce I/O bottlenecks. To mitigate this:

- Tile saving was buffered.
- Immediate deletion of in-memory PIL objects was enforced.

2. JPEG Compression Trade-off Tiles are saved as JPEG (quality=90) rather than PNG to reduce storage footprint by 60%. Empirical validation confirmed negligible degradation in segmentation while significantly improving disk throughput.

3. Coordinate Traceability Without strict spatial bookkeeping, reconstruction would be impossible. The manifest.csv was therefore designed as a deterministic mapping layer between spatial coordinates and extracted tiles, enabling reproducible full-slide reconstruction.

4 Module B: Deep Learning Segmentation

4.1 Implementation Strategy

The training pipeline was built using PyTorch and `segmentation-models-pytorch`.

- **Augmentation:** To handle H&E staining variability (purple/pink shifts), we applied aggressive `ColorJitter` (Brightness=0.2, Contrast=0.2, Saturation=0.2) and geometric augmentations (Flip, Rotate).
- **Loss Function:** **Dice Loss** was chosen over Cross-Entropy to address the class imbalance between tissue (foreground) and space (background).

4.2 Loss Function Justification

Class imbalance in histopathology segmentation is severe. Let:

P = number of positive pixels

N = number of negative pixels

Typically, $P \ll N$. Cross-Entropy loss tends to bias toward predicting background due to the dominance of negative samples.

Dice Loss directly optimizes the overlap metric:

$$Dice = \frac{2|A \cap B|}{|A| + |B|}$$

Its gradient formulation inherently balances foreground and background contributions, making it more suitable for sparse-object segmentation tasks.

Empirical comparison showed:

- Cross-Entropy: Faster early convergence but plateaued at lower IoU.
- Dice Loss: Slower initial training but superior final boundary delineation.

4.3 Training Metrics Convergence

The model was trained for up to 100 epochs. An **Early Stopping** mechanism (Patience = 15) was implemented to prevent overfitting. As seen in Table 3, the model reached its peak performance at Epoch 66.

Table 3: Training History Highlights (from `history.json`)

Epoch	Train Loss	Val Dice	Val IoU	Status
1	0.6033	0.3871	0.2443	Initial
10	0.3521	0.6542	0.4912	Improving
30	0.1845	0.7821	0.6433	Converging
60	0.1321	0.8011	0.6715	Plateau
66	0.1264	0.8134	0.6872	Best Model

4.4 Final Validation Performance

The final model performance, as recorded in `val_metrics.json`, confirms the efficacy of the chosen architecture:

- **Architecture:** ResNet50-UNet (Heavy Finetune)
- **Validation Dice Score: 0.8134**
- **Validation IoU: 0.6872**
- **Final Training Loss: 0.1264**



Figure 2: Training Dynamics. The curves demonstrate stable convergence, with the validation metrics (IoU/Dice) correlating well with the reduction in training loss.

5 Module C: Inference Artifact Filtering

5.1 The Efficiency Challenge

A typical WSI is composed of $> 80\%$ empty glass background. Processing these white tiles is computationally wasteful and can lead to false positives (artifacts being detected as tissue).

5.2 Evolution of the Smart Filter

We iterated through several methods to filter out background tiles:

1. **Luminance Thresholding:** Initially, we filtered tiles with mean brightness > 230 . This failed due to vignetting (dark corners) and dust artifacts, which were dark but contained no tissue.

2. **Final Solution: Statistical Variance Filter.** We implemented a filter based on Standard Deviation (σ).

$$\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2} \quad (1)$$

Logic: Empty glass (flat white) and solid artifacts (flat black) have near-zero variance. Cellular tissue is highly textured, resulting in high variance. A threshold of $\sigma > 5$ proved robust, reducing inference time by approximately 70%.

5.3 Statistical Foundation of the Variance-Based Filter

Let X be the grayscale intensity random variable of a tile. For empty glass regions:

$$Var(X) \approx 0$$

For tissue regions, due to cellular nuclei, cytoplasm texture, and staining heterogeneity:

$$Var(X) \gg 0$$

Empirical histogram analysis across 500 sampled tiles revealed:

- Empty regions: $\sigma \in [0, 3]$
- Tissue regions: $\sigma \in [12, 45]$

A conservative threshold of $\sigma > 5$ was selected to minimize false negatives while filtering most background regions.

This reduced inference computations by approximately 70%, without measurable degradation in segmentation recall.

5.4 Sliding Window Inference

To perform inference across the whole slide, the system employs a sliding-window approach with a defined overlap (e.g., 64px). This overlap is crucial to prevent "checkerboard artifacts" at the edges of the tiles, where the model lacks context.

5.5 Demo Mode Heuristic

To enable instant demonstrations without scanning the entire slide, a heuristic search algorithm was implemented. This algorithm analyzes the slide's low-resolution thumbnail to generate an entropy map, automatically identifying the region with the highest tissue density to initiate the inference process.

5.6 Final Inference Result

Figure 3 displays the final segmentation mask generated by the system on a test tile.



Figure 3: Final Inference Output (**sample**). The model successfully segments the tissue region (white) while ignoring the background (black), demonstrating precise boundary delineation.

6 Conclusion

HistoScan AI demonstrates that high-resolution histopathology segmentation is achievable on consumer-grade hardware through carefully engineered architectural compromises.

The key innovation lies not in model novelty, but in system-level optimization:

- Native-resolution processing without destructive resizing
- Variance-based intelligent tile filtering
- Memory-aware architectural selection
- Robust training stabilization mechanisms

The final validation Dice score of 0.8134 confirms that preserving spatial resolution was a justified decision despite reduced batch size.